

Proposal Training

GDG PUP '26 Cybersecurity Team – Cybersecurity Hands-on Online Training Resource

Overview

These initiatives combine practical training, gamified learning, and hands-on challenges to strengthen both foundational and advanced cybersecurity skills.

1. Cybersecurity Educational Website (with Sign-in)

Description

A **centralized cybersecurity practice platform** where members can safely run scripts, test vulnerabilities, and practice real-world attack/defense techniques. The platform will feature isolated “practice rooms” designed for common web vulnerabilities and will include CTF-style flags to guide learning.

Key Features

- **Safe Vulnerability Practice Rooms**
Dedicated mini-web applications purposely designed with vulnerabilities (e.g., XSS, SQLi, IDOR). + OSINT
Members can test payloads, experiment with scripts (makikita effect nung code), and explore how attacks work.
- **CTF-Style Challenges With Hidden Flags**
Each vulnerable module includes hidden flags that participants must discover through correct exploitation techniques.
Flags will be unique to each module and act as “proof of learning.”
- **Google Classroom Integration**
Members submit their captured flags directly through Google Classroom for tracking and grading. (Special Awards: Top 10)
- **Centralized Learning Experience**
Eliminates the need for external tools or environments.
All content, challenges, and practice labs are hosted in one GDG-managed website.

Objectives

- Provide a safe, controlled environment for hands-on cybersecurity training.
- Build foundational and intermediate skills in web security exploitation.
- Encourage self-paced learning and structured progression through modules.

2. GDG Cybersecurity Discord Bot + Daily CTF System

A **Discord bot** that delivers scheduled cybersecurity mini-CTF challenges to the GDG community. These will be short, bite-sized tasks modeled after “Wordle-style” daily puzzles, ensuring consistent practice and community engagement.

Key Features

- **Daily or Bi-Daily Cybersecurity Challenges - Ticket**

Challenges can be sourced from:

- Internal Google CTF challenges
- Public write-ups on GitHub
- Simplified real-world vulnerability cases

Each challenge will be short, solvable within minutes, and focused on a specific skill.

- **Automated Scoring System**

Responses are submitted privately to the bot.

The first solver receives the highest points, with decreasing points for later solvers.

- **Community Leaderboard**

A dynamic leaderboard ranks members based on accumulated points.

This encourages consistent participation and friendly competition.

- **Solution Reveal After Challenge Closure**

Once a challenge expires, the bot automatically posts the official solution, allowing participants to learn and review correct methods.

- **CTF Season Format (Optional)**

Points carry over into “seasons” (e.g., monthly or per semester). - In terms of topic

Top performers may receive recognition, badges, or rewards.

Objectives

- Maintain consistent member engagement through gamified learning.
- Strengthen analytical and problem-solving skills via short CTF exercises.
- Build a community-driven culture of collaboration, learning, and improvement.

Website

Cybersecurity Educational Website

A Guided, Hands-On Learning Platform for Red & Blue Teams

Purpose

A **study-first cybersecurity learning resource** where members safely practice Red and Blue Team skills at their own pace.

No competition. No time pressure. Learning is validated through simple **checkpoints (flags)**.

User Flow (TryHackMe-like Experience)

1 Onboard

- User logs in → lands on **Learning Dashboard**
- Sees Sessions 5–12 as a **guided roadmap**
- Progress is **private and self-paced**

2 Choose a Session

- Red Team sessions (Web, Post-Exploitation, Cloud Attacks)
- Blue Team sessions (Hardening, SIEM, Incident Response)
- Each session explains:
 - *What you'll learn*
 - *Why it matters in real life*

3 Enter a Room

- Clear instructions (what to read, what to try)
- External resources linked:
 - PortSwigger

- OWASP Juice Shop
- Official docs (Burp, Wazuh, GCP)
- Practice environment provided (DVWA, VM, logs, cloud sim)

4 Practice & Explore

- Users experiment freely
- No penalties, no timers
- Encouraged to observe, break, and understand

5 Learning Checkpoint

- One simple flag confirms understanding
- Points mark **completion only**, not ranking
- Room marked “Completed”

6 Progress Forward

- Next room/session unlocks
- Builds confidence toward **SEEN: Cyber Rumble**

Red Team Learning Path

- Understand vulnerabilities (SQLi, XSS, auth bypass)
 - Learn post-exploitation and privilege escalation
 - Explore cloud misconfigurations safely
 - Focus: *How attacks work and why they succeed*
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Blue Team Learning Path

- Apply baseline hardening and secure configs
 - Learn to read logs and detect attacks
 - Practice incident response decision-making
 - Focus: *How defenses prevent or detect attacks*
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How to Develop This Platform (High-Level)

- **Frontend:** Simple room-based UI (TryHackMe style)
- **Backend:**
 - Room unlock logic
 - Flag validation
 - Progress tracking
- **Labs:**
 - DVWA / Juice Shop
 - Downloadable VMs
 - Log files & cloud simulators
- **Design Principle:**
Learning-first, low-pressure, expandable every semester

User Flow

Revised User Flow — Cybersecurity Educational Website (Study-First)

Core Philosophy

- Learning > competition
- Exploration > speed
- Guidance > pressure
- Flags = **learning checkpoints**, not rankings

Points exist only to **mark completion**, not to compare students.

0. Entry Point

User logs in with their GDG account.

No timers, no public leaderboard.

They are welcomed with:

“This platform is a guided space to practice what you learn in our sessions. Take your time.”

1. Home Dashboard (Learning Hub)

What the user sees:

- List of **Sessions 5–12**
- Each session marked as:

- Not Started
 - In Progress
 - Completed
- Personal progress indicator (private)

Example:

✓ Session 5: Web Exploitation
○ Session 6: Host Hardening
🔒 Session 9: Post-Exploitation (opens after Session 6)

No scores shown publicly.
Progress is **for the learner only**.

2. Session Page (Study Modules)

Clicking a session opens a **study-focused overview page**.

Example: Session 5 — Web Exploitation

The page includes:

- Session goal (plain language)
- What this skill is used for in real life
- List of learning rooms
- Recommended time (flexible, not enforced)

Example text:

“This session introduces common web vulnerabilities and how attackers exploit them. The goal is understanding—not speed.”

3. Room Page (Guided Practice Room)

Each room feels like a **guided worksheet with a sandbox**.

A. Instructions (Main Focus)

Clear and calm structure:

- What concept this room teaches
- What to read/watch first
- What to try afterward

Example:

Before you start:

Read the PortSwigger guide on SQL Injection (Basics).

Then:

Explore the vulnerable login form and observe how input is handled.

No “hack this fast” language.

B. External Learning Resources

Each room links to:

- PortSwigger Academy
- OWASP Juice Shop documentation
- Official tool docs (Burp, Wazuh, GCP)

These are framed as **study references**, not prerequisites.

C. Practice Area

- Link to DVWA / Juice Shop / VM / simulated cloud
- Tool suggestions (optional, not enforced)
- Encouragement to experiment safely

Text example:

“Try different inputs and observe how the application responds. You are encouraged to break things in this controlled environment.”

D. Learning Checkpoint (Flag Submission)

Instead of “Capture the Flag”, call it:

Learning Checkpoint

- One simple flag per room
- Flag confirms understanding (e.g., finding a vulnerability, identifying a log, securing a setting)
- Unlimited attempts
- No penalties

Text example:

“Submit the flag once you’ve completed the exercise. This confirms that you explored the concept.”

E. Completion Feedback

When submitted:

- Friendly confirmation message
- Room marked as completed

- Optional reflection prompt:
 - “What did you notice?”
 - “What confused you?”

No ranking. No pressure.

Red Team Session Flow (Study-Oriented)

Session 5 — Web Exploitation

- Explore SQLi, XSS, auth bypass
- Understand *why* it works
- Flag proves observation, not mastery

Session 9 — Post-Exploitation

- Learn what privilege escalation looks like
- Understand risk and impact
- Flag confirms successful exploration

Session 11 — Cloud Attacks

- Understand misconfiguration risks
 - See how cloud mistakes happen
 - Flag shows awareness, not exploitation speed
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Blue Team Session Flow (Study-Oriented)

Session 6 — Host Hardening

- Learn baseline security steps
- Apply common protections
- Flag confirms system reached a safer state

Session 10 — IDS & SIEM

- Learn what logs and alerts look like
- Recognize patterns, not write perfect rules
- Flag confirms correct identification

Session 12 — Incident Response

- Learn IR lifecycle
- Practice thinking through response steps
- Flag confirms correct decision-making